EW Overview

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Objective

understanding of electronic warfare (EW) components/functions

Learning Outcomes

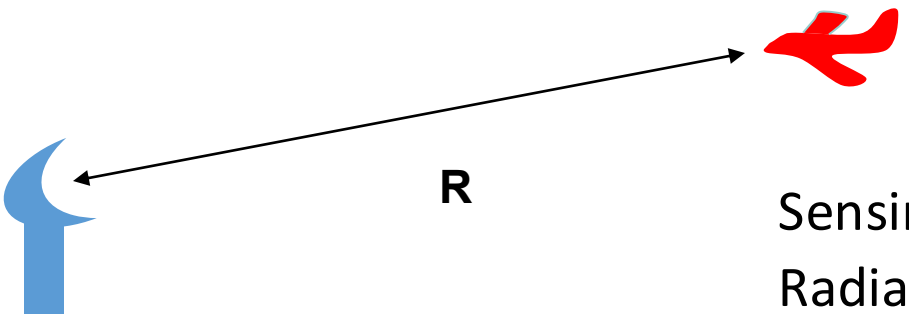
At the end of the lesson, you should be able to

1. describe EW components/functions
2. discuss EW components and their relationships
3. describe functions of ELINT, COMINT and SIGINT
4. discuss C4ISR, and how it is related to EW components

Presentation Plan

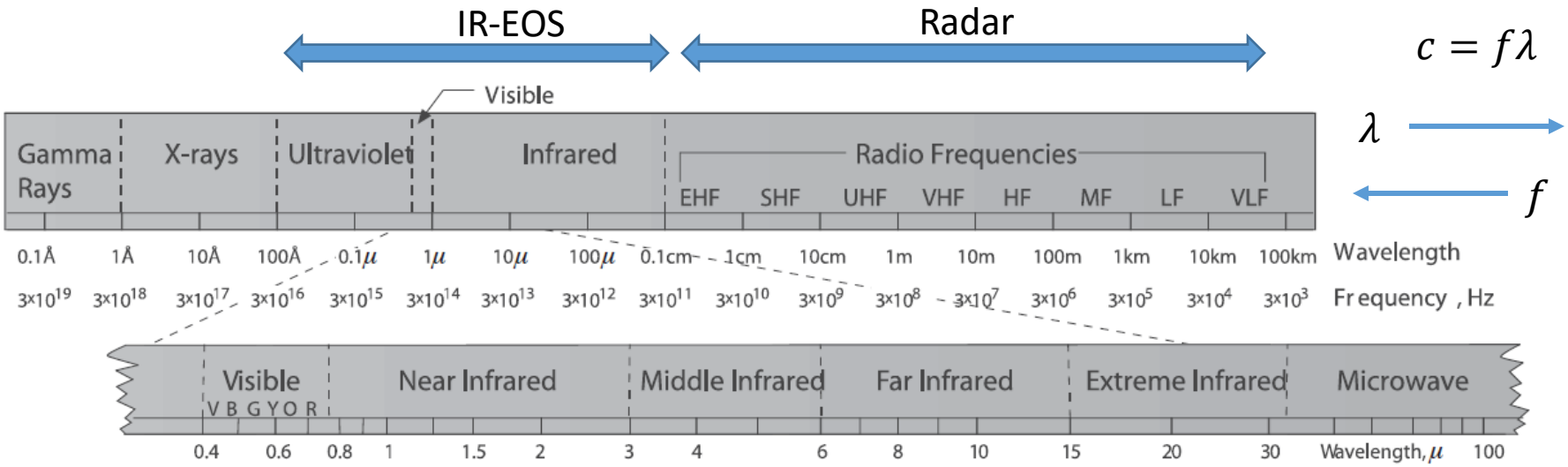
- EW Overview
- Functions/Components of EW
  - Electronic Support (ES)
  - Electronic Attack (EA) (ECM)
  - Electronic Protection (EP) (ECCM)
- ELINT/COMINT/SIGINT/C4ISR

Sensors (emitting energy->emitter)



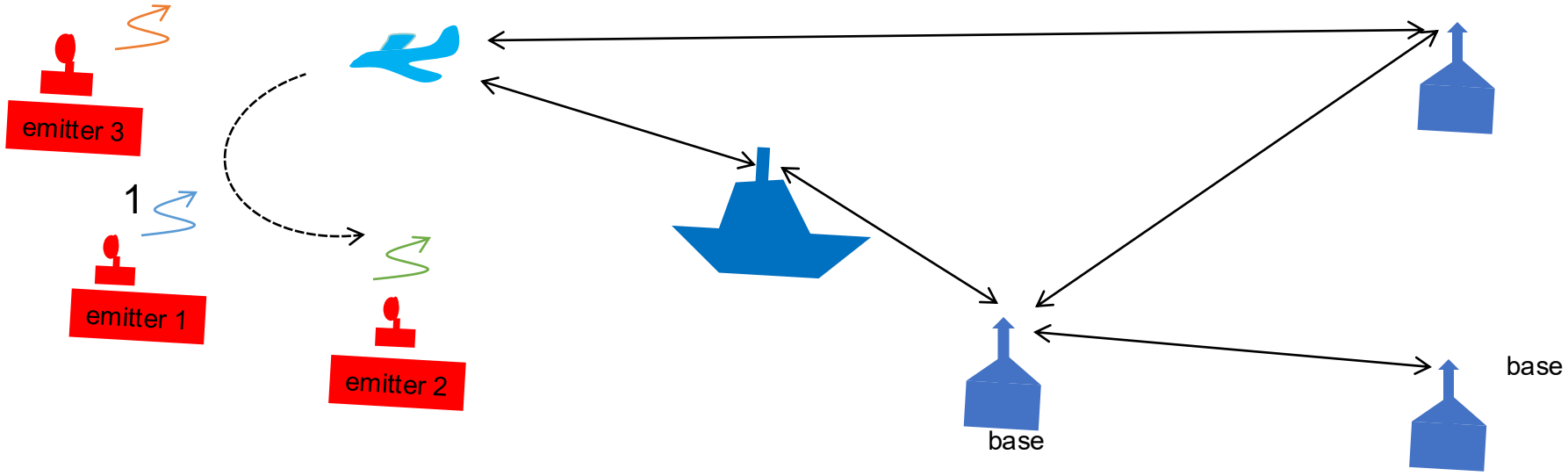
Sensing system (radar, laser, sonar etc.)

Radiation of energy (RF, light or sound)



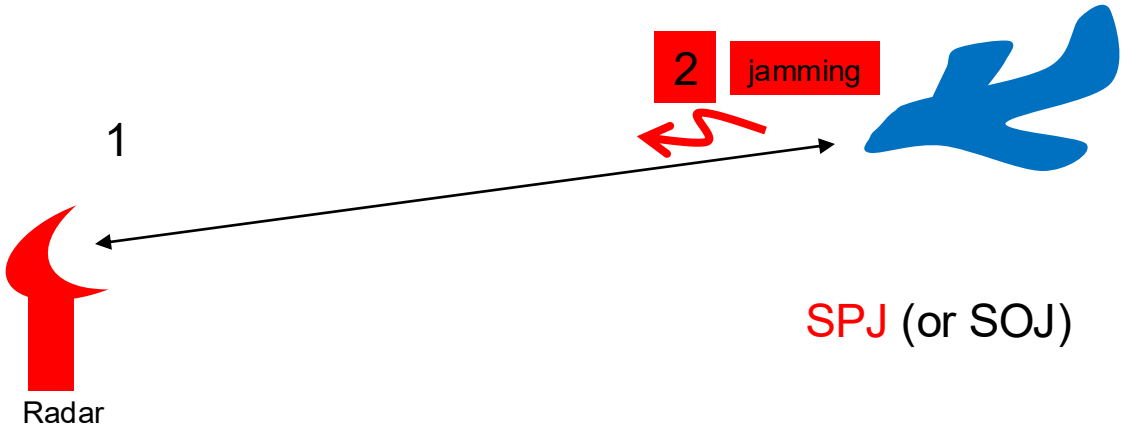
Electronic Support Measures (ES) (or ESM)

- Actions involving intercepting, identifying and locating of EM energy
- Serves to EP and/or EA, try to recognise threats for EP or EA
- Also known as Electronic Support Measures (ESM)
- involves operational factors of enemy's equipments



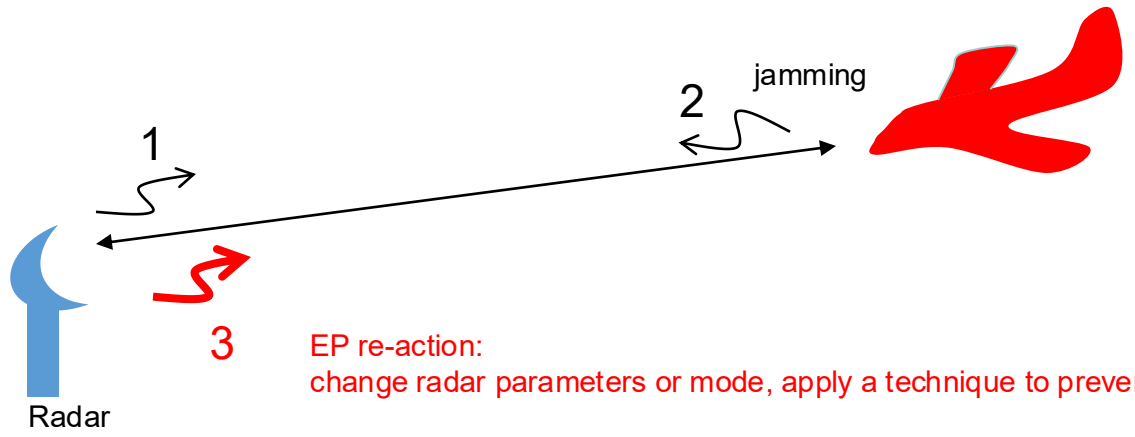
Electronic Attack (EA)

- Any process to prevent operation of detection systems such as radar, laser, sonar etc.
- Previously known as Electronic Counter Measures (ECM)
- “Soft kill” or “hard kill”



Electronic Protection (EP)

- Any process to prevent operation of EA (ECM) systems
- Previously known as Electronic Protective Measures(EPM)
- Also known as Electronic Counter-Counter Measures(ECCM)



Electronic Warfare (EW) [Martino]

formal military terminology -> *electronic warfare* (EW) as a military action whose objective is the **control of the *electromagnetic spectrum*** (EMS)

EW Functions (components)

- Electronic support (ES) (near real time threat recognition)
- Electronic attack (EA) (ECM- Electronic Counter Measures- ”Soft Kill”)
- Electronic protection (EP) (ECCM- Electronic Counter-Counter Measures)

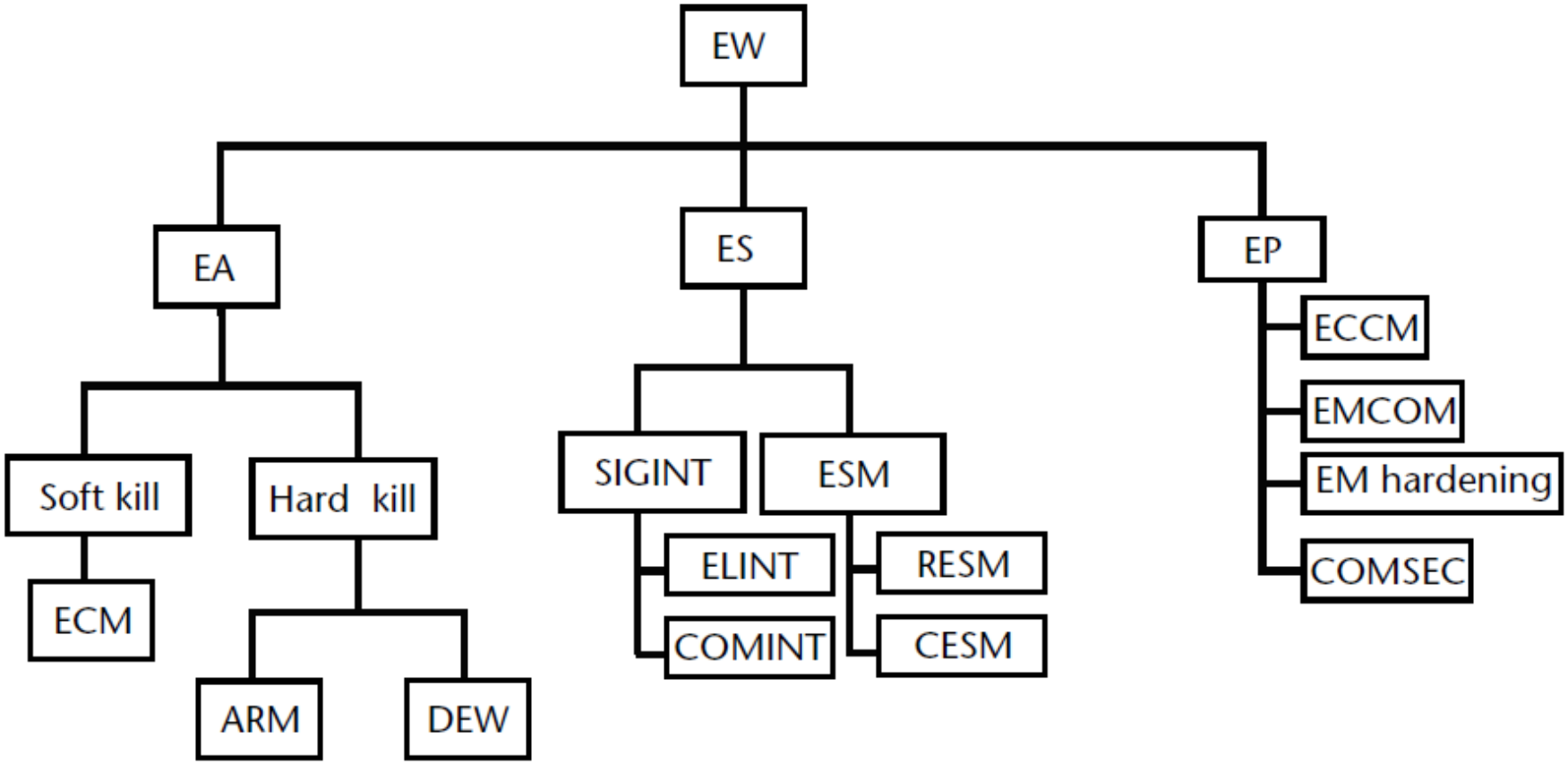
Electronic Attack (EA) [hard kill]

- directed-energy weapons (DEWs)
- antiradiation missiles (ARMs)

Electronic Protection (EP)

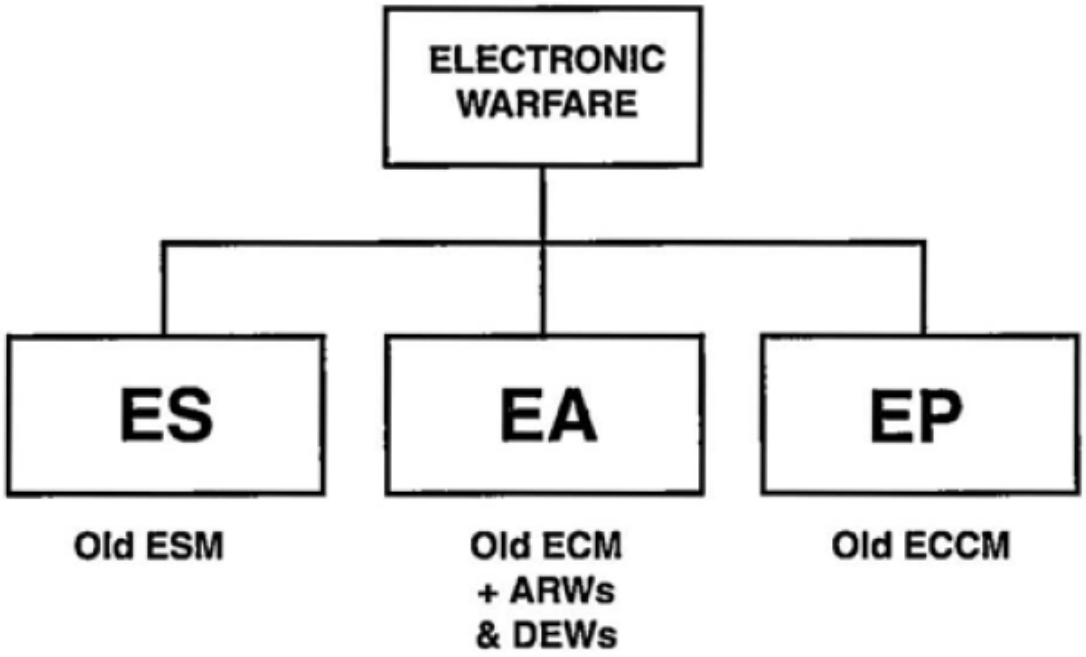
- electromagnetic emissions control (EMCON)
- electromagnetic hardening of the electronic equipment (EMC)
- communications security (COMSEC) actions

EW Functions [Martino]



EW [Adamy]

Electronic warfare -> *the art and science of preserving the use of the electromagnetic spectrum for friendly use while denying its use to the enemy*



Information Warfare (IW) and EW [Adamy]

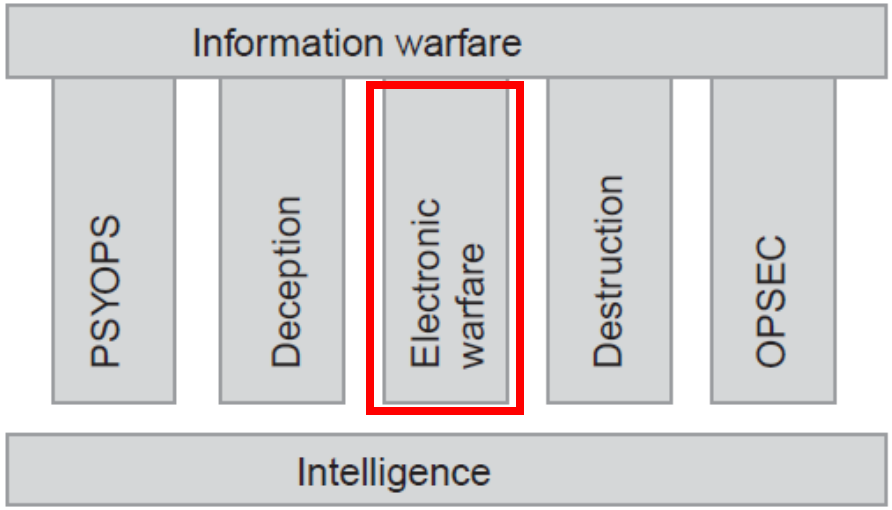
- *actions taken to preserve the integrity of one's own information system from exploitation, corruption, or disruption,*

while, at the same time,

- *exploiting, corrupting, or destroying an adversary's information system,*

as well as

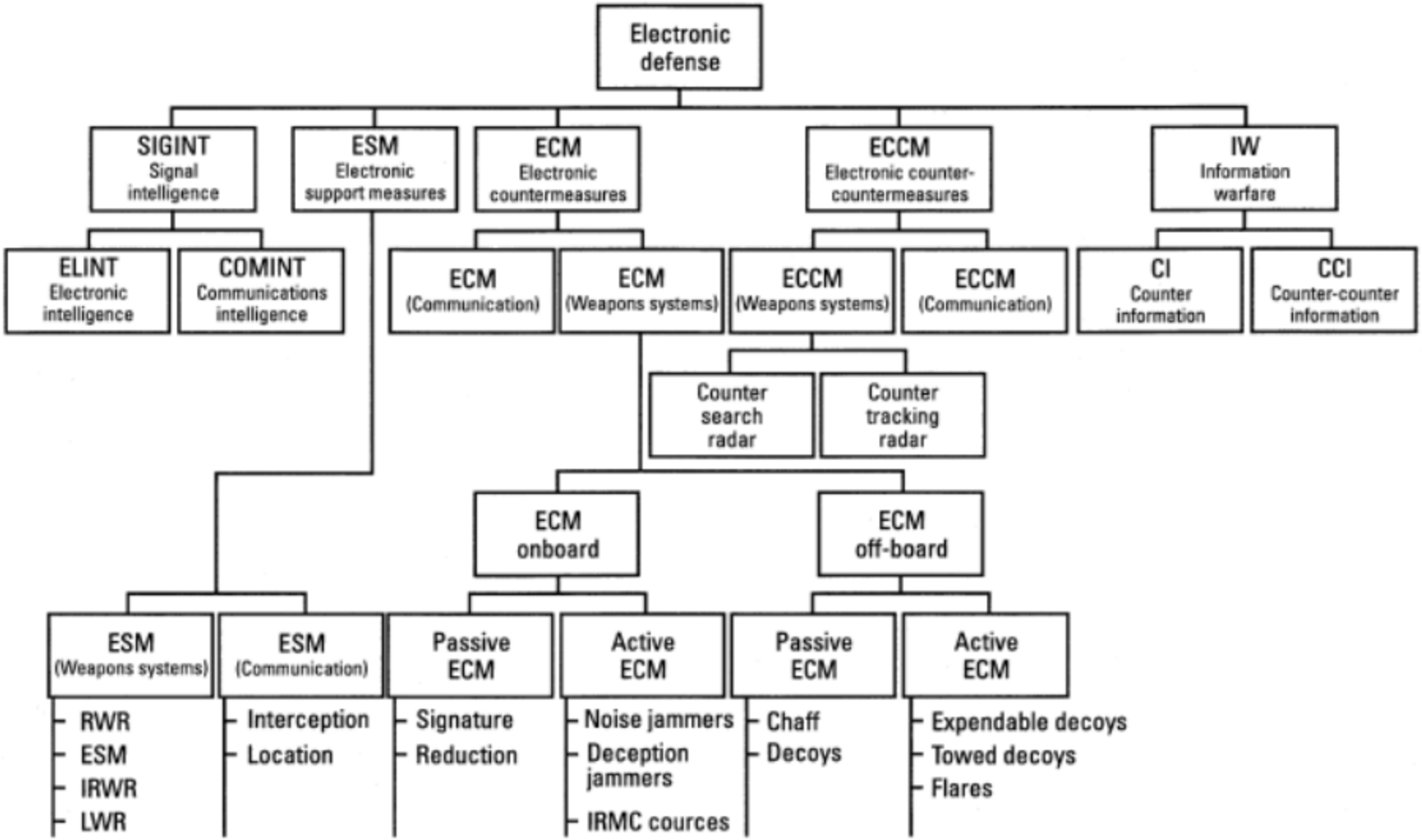
- *the process of achieving an information advantage in the application of forces*



Pillars of Information Warfare (IW)

- EW-> integral part of IW (action part)
- psychological operations (PSYOPS)
- deception
- physical destruction
- operational security (OPSEC)

Electronic Defense [Neri]



SIGINT/ELINT/COMINT

Communications Intelligence(COMINT):

- intercepting and analysis of communication emissions
- identification of communications networks

Electronic Intelligence(ELINT):

- intercepting and analysis of emissions from electronic equipments(emitter) such as radars, sonars, lasers etc.
- identify type of emission systems, electronic parameters, configurations

Signal Intelligence(SIGINT)

- Acquisition of data about EM emissions of potential enemy, knowledge of enemy electronic systems (warfare and communications)
- Specification (or exploring) of capabilities of weapon systems
- The strategy the enemy will use in the employment of weapon systems
  Example: SAM and AAA sites, location-capability
- ELINT and COMINT -> SIGINT

- ESM, ELINT, COMINT-> receiving enemy transmissions!

SAM: surface to air missile AAA: Anti-aircraft artillery

C4ISR

Command, Control,Communications, Computers, Intelligence, Surveillance and Reconnaissance(C4ISR)

Group of functions to enable the coordination of operation

- Command and Control is the ability of commander to direct his forces using communications based on the intelligence (surveillance & reconnaissance) acquired
- Strategical or tactical

How they(!) describe C4ISR

- turning data into knowledge, and knowledge into action [Lockheed Martin]
- capabilities provide an advantage through situational awareness, knowledge of the adversary and environment, and shortening the time between sensing and response [Northrop Grumman]
- C4ISR Development Program yielded its products under the categories of 1) Fire Support (FS) 2) Air Defense (AD) 3) Battle Management System (BMS) 4) Electronic Warfare (EW) 5) Reconnaissance/Surveillance and Intelligence and 6)Tactical Communications Systems and Equipment [Aselsan]